

LC 38: Count and Say

Wells Fargo

Zoho

PROBLEM NOTES

- just an `n` will be given
- `cands(1) = "1"`
- `cands(n) = run length encoding of cands(n-1)`
- RLE of `11` = `"21"`

DRY RUN / ITERATIONS (LET N=5)

Iteration	Operation	Result (ans)
1st iteration	-	1
2nd iteration	RLE(1) = "11"	11
3rd iteration	RLE(11) = "21"	21
4th iteration	RLE(21) = "1211"	1211
5th iteration	RLE(1211) = "111221"	111221

APPROACH

since n is small, precompute for all

```
ans[0] = "1"
```

For RLE function → maintain idx pointer

```
if(s[idx] == s[idx+1]) { idx++; pref++; } else { ans += pref ans += s[idx]
```

C++ IMPLEMENTATION

```
class Solution { public: string rle(string s){ int idx=0; int pref=1;
char subject=s[idx]; int n=s.size(); string ans=""; while(idx<n-1){
if(s[idx]==s[idx+1]){ idx++; pref++; } else{ ans+=to_string(pref);
ans+=subject; idx++; //reset pref=1; subject=s[idx]; } } ans+=to_string(pref);
ans+=subject; return ans; } string countAndSay(int n) { vector<string>
cants(30); cans[0]="1"; for(int i=1;i<=29;i++){ cans[i]=rle(cans[i-1]); }
return cans[n-1]; } };
```